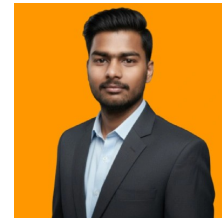


Lekkala Ganesh

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PROFESSIONAL SUMMARY

Full-stack developer with **3 professional roles** across AI-powered products, healthcare UI, and scalable web platforms. Skilled in **LLM orchestration**, multi-provider fallback, **VRAM optimization**, and building accessible (**WCAG 2.1 AA**), secure (**RLS**, **JWT**) products with React, Next.js, and FastAPI. Seeking software engineering roles in Bangalore.

TECHNICAL SKILLS

AI & ML: PyTorch, Hugging Face (Diffusers, Transformers), 4-bit Quantization (bitsandbytes), CUDA, Scikit-learn, LLM API Integration (GPT-4o, Gemini), Model Fallback Architectures
Enterprise & JVM: Java 21+, Spring Boot, REST APIs, Angular (v18+), TypeScript
Backend & Data: REST/WebSocket APIs, FastAPI, Node.js, PostgreSQL (Row-Level Security), Supabase, MongoDB Atlas
Frontend: React, Next.js (SSR/SSG), TypeScript, Tailwind CSS, Framer Motion, Three.js
DevOps & Tools: GitHub Copilot (AI-native development), Docker, GitHub Actions (CI/CD), Linux Shell Scripting, Vercel, Git
Testing & Accessibility: Vitest, Playwright (E2E), WCAG 2.1 AA, SEO (JSON-LD)

PROFESSIONAL EXPERIENCE

Freelance Full-Stack Developer

Self-Employed

Remote

Jan 2025 – Present

- **End-to-End Ownership:** Owned full-stack delivery end to end for multiple client-facing web apps — designing and integrating **REST APIs** and re-architecting legacy front-ends into modern component-driven UIs — shipping each from requirements to production with **rapid turnaround and zero critical bugs**.
- **AI System Integration:** Engineered a **multi-provider AI fallback** (GPT-4o, Gemini, OpenRouter) with graceful degradation, powering real-time SSE voice streaming ([VoxAI](#)) and 3D shadow simulations ([SunScope Pro](#)).
- **Healthcare UI & Secure Backends:** Built **12+ modular React components** for a patient-facing portal (**40%** fewer accessibility findings, HIPAA-aligned), with **RLS + JWT** secured backends on Supabase/PostgreSQL for [Finance Tracker](#) and [2A Construction](#).
- **Accessibility & Cross-Browser:** Achieved **WCAG 2.1 AA** via custom ally hooks and automated Playwright testing; delivered **100% visual parity** across Chrome, Firefox, Safari, and Edge.

Full Stack Web Development Intern

Slash Mark IT Solutions

Remote, Part-time

Dec 2024 – Apr 2025

- Delivered **6 full-stack projects** across 3 tiers: real-time Weather Forecast, URL Shortener API (Node.js/Express/MongoDB), and a Food Delivery platform with auth, geolocation, and payment flow.

Machine Learning Intern

Bharat Intern

Virtual

Oct 2023 – Nov 2023

- Built regression/classification models (House Price, Wine Quality, Iris) with **Scikit-learn** and Pandas, applying feature engineering, cross-validation, and hyperparameter tuning.

KEY TECHNICAL PROJECTS

AuraGen – AI Image & Video Generation Platform | PyTorch, HF Diffusers, FastAPI, Next.js, CUDA

[GitHub](#)

- **VRAM Optimization:** Engineered a **4GB VRAM** stack (NF4 quantization, attention slicing, VAE tiling) cutting model memory from **12.8GB to 3.2GB**, enabling text-to-video (Wan 2.1) on consumer GPUs.
- **System Orchestration:** Designed a real-time job queue with WebSocket progress streaming and **hybrid cloud burst routing** (HuggingFace Inference fallback) for peak loads.

GoVault – AI-Powered Travel Organizer | Next.js, TypeScript, Supabase, GPT-4o, Playwright

[GitHub](#)

- **AI-Powered Import:** Built a **GPT-4o** integration extracting structured booking data from emails/receipts, with **offline-first** architecture (IndexedDB + background sync).
- **Full Travel Suite:** Implemented itinerary builder, multi-currency expense tracking, traveler profiles, and collaborative trip sharing with privacy controls.

SymptomChecker AI – ML-Powered Health Assistant | Flask, Scikit-learn, React, MongoDB, JWT

[GitHub](#)

- **Model Accuracy:** Built a Scikit-learn prediction engine achieving **95%+ accuracy across 40+ medical conditions**, integrated with location-based hospital recommendation APIs.

EDUCATION

Parul University

Bachelor of Technology in Computer Science & Engineering; GPA: 7.53/10

Vadodara, Gujarat

Jul 2022 – Jun 2025